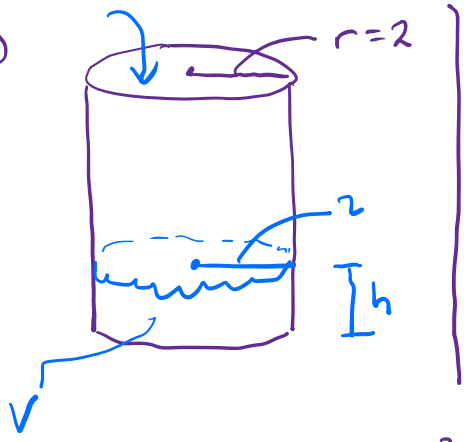
Problem 3. WORKED IN CLASS

A cylindrical tank of radius 2 m is being filled with water at 3 cubic meters per minute. How fast is the **water level** rising?

1. Draw + Label
2. What & when
3. Equation
4. Differentiate w.r.t. t
5. Sub + Solve

(1)



(2) $\frac{dV}{dt} = 3$

$\frac{dh}{dt} = ?$

(3) $V = \pi r^2 h = \pi 2^2 \cdot h = 4\pi h$

(4) $\frac{d}{dt} V = \frac{d}{dt} 4\pi h$

$\frac{dV}{dt} = 4\pi \frac{dh}{dt}$

(5) $3 = 4\pi \frac{dh}{dt}$

$\rightarrow \frac{dh}{dt} = \frac{3}{4\pi} \frac{\text{meters}}{\text{minute}}$